Jiajun Liu

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Basic Information

- **Education:** *Bachelor's degree* (2023 - 2027), **Gaoling School of Artificial Intelligence, Renmin University of China**
- **Academic Performance: GPA: 3.88/4.00 (Rank 3 / 58)**
- **Research Interest: Spatial Intelligence, World Model, Embodied AI**
- **Language Proficiency: Mandarin (Native), English (Fluent speaker, CET-6 Test Score: 604; Oral: Excellent)**
- **Professional Skills: Linux development, Python, C/C++, PyTorch**

Research Experiences

Research-Intern, IIIS, Tsinghua University, China Oct 2025 - Present
- **Advisor**: Mengdi Xu
- **Topic**: Robot learning with few-shot demonstrations and non-expert data & world action models

Research Intern, MLL-Lab, Northwestern University, USA Jun 2025 - Sep 2025
- **Advisor**: Manling Li
- **Topic**: VLM agents with tool-use for active novel-view synthesis on spatial intelligence tasks, in collaboration with Microsoft.

Research Intern, GSAI-ML Group, Renmin University, China Jan 2025 - Aug 2025
- **Advisor**: Chongxuan Li Jun Zhu
- **Topic**: 3D world model for explorable scene generation from a single image or text prompt & interactive video world model.

Publications & Preprints

1. **Jiajun Liu***, Yifan Chen*, Yichao Liu*, Jiayi Zhang, Ruoqu Chen, Shaoxuan Xie, Guocai Yao, Mengdi Xu, Sen Cui, Changshui Zhang. RoboCoach: World Models as Active Coaches for Compositional Robot Skills. *Preprint*, 2026.
2. **Jiajun Liu**, Jieming Li, Zi Zhuang, Hang Yu, Qingli Chen, Liu Cao, Yingxi Lu, Ruoqu Chen, Yuhang Cao, Chenyu Zhang, Yankai Lin, Mengdi Xu. What Do VLAs Actually Learn through In-Context Failure Conditioning? *CVPR@3D-LLM/VLA*, 2026.
3. Chendong Xiang*, **Jiajun Liu***, Jintao Zhang*, Xiao Yang, Zhengwei Fang, Shizun Wang, Zijun Wang, Yingtian Zou, Hang Su, Jun Zhu. Geometry-Aware Rotary Position Embedding for Consistent Video World Model. *ICLR@WorldModels*, 2026.
4. Liu Cao, Botian Xu, Ruoqu Chen, **Jiajun Liu**, Mingzhi Pei, Zihao Wang, Zhongyu Li, Mengdi Xu. HuMiT: Low-Latency Whole-Body Humanoid Teleoperation via Minimal Reference Tracking. *Preprint*, 2026.
5. Ruoqu Chen, Feixiang Ruan, Liu Cao, Zihao Wang, Botian Xu, Shiqin Tong, **Jiajun Liu**, Wanli Xing, Kaifeng Zhang, Mengdi Xu. Dex-X: Learning Visual-Tactile Dexterous Manipulation From Human Demonstration with Simulated Interaction. *CoRL*, 2026; *RSS@Dex* (Spotlight), 2026.
6. Chenyu Zhang*, Yuhang Cao*, Daru Du, Yingxi Lu, Jing Shao, Ruoqu Chen, **Jiajun Liu**, Liu Cao, Yicheng Liu, Hang Zhao, Mengdi Xu. Rethinking Causal Action Tokenization with Conditional Annealing in Flow Matching. *NeurIPS*, 2026; *NeurIPS@WRL* (Oral), 2026.
7. Wenqi Zhang*, Mengna Wang*, Gangao Liu, Huixin Xu, Yiwei Jiang, Yongliang Shen, Guiyang Hou, Zhe Zheng, Hang Zhang, Xin Li, **Jiajun Liu**, Weiming Lu, Peng Li, Yueting Zhuang. Embodied Reasoner: Synergizing Visual Search, Reasoning, and Action for Embodied Interactive Tasks. *ACL*, 2026.

Open-Source Projects

- **Embodied Reasoner (With OSPP's funding, 260+ Github Stars)** *Main contributor*
  - Embodied-Reasoner (ER.) is a multimodal model designed for deep reasoning and long-horizon interaction. Through OSPP, similar to GSoC, I was selected by the AGIROS Community as the contributor responsible for ER. I took charge of benchmarking ER. on ALFRED and contributed to resolving two key bottlenecks: ambiguity in distinguishing identical object instances and imprecise targeting of large objects, thereby improving spatial accuracy and interaction robustness.
- **RAGEN & VAGEN: Training Agents by Reinforcing Reasoning (With 2.8k+ Github Stars)** *Contributor*
  - This twin of projects empower agents with RL to operate effectively in interactive and stochastic environments by handling multi-turn interactions and environmental uncertainty. I contributed to developing more environments and mask functions to compute the loss only for the parts generated by the model, which actually made training more stable.

Selected Awards

- **Silver Medal, International Collegiate Programming Contest (ICPC) East-Asia Continent Final** Dec 2024
  Awarded by ICPC Foundation
- **Gold Medal, International Collegiate Programming Contest (ICPC) Asia Continent Regional (Wuhan)** Nov 2025
  Awarded by ICPC Foundation
- **Gold Medal, "Xiaomi Cup" China Collegiate Programming Contest (CCPC) Invitational Contest** Apr 2025
  Awarded by CCPC Committee and Xiaomi (Rank 8-th)
- **Silver Medal, 2024 CCF Collegiate Computer Systems & Programming Contest (CCSP)** Oct 2024
  Awarded by China Computer Federation (CCF)
- **Most Potential Award, Open Source Promotion Plan (OSPP) 2025** Dec 2025
  Awarded by Institute of Software, Chinese Academy of Sciences (ISCAS) and Huawei (4 Nationwide)

Selected Scholarships

- **National Scholarship** Sep 2025
  Awarded by Ministry of Education of the People's Republic of China (¥10,000, Top 0.4% Nationwide)
- **"Linghang" Intellectual Excellence Dean's Scholarship** Dec 2024
  Awarded by ByteDance and Gaoling School of Artificial Intelligence (¥5,000)
- **China National Petroleum Corporation Scholarship** Sep 2026
  Awarded by Renmin University and China National Petroleum Corporation (¥12,000)
- **SenseTime Scholarship Nomination Award** Jun 2026
  Awarded by SenseTime (Top 40 Nationwide)

Services & Presentations

- **Conference Reviewer:** ICLR 2027
- **Video: "My Experience Using AI+ Tools to Create Videos"** Oct 2024
  Invited by China Computer Federation (CCF) for *China National Computer Conference (CNCC) Super Forum*
- **Talk: "Algorithm and Artificial Intelligence"** Mar 2025
  Invited by Gaoling School of Artificial Intelligence & School of Information
- **Talk: "Robot Learning from Failure and Physical RSI"** 6Oct 202
  Invited by TSAIL-Group and BigEye-Lab